Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Developing a Culturally Responsive Socially Assistive Robot for Emotional Care of Adolescents in Multicultural Environments**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication (√), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Robot & Intelligent Systems		*Robot & Intelligent Systems*	
	Medium Category	Small Category	Medium Category	Small Category
	Service Robots	*Social Interaction Robots / Emotional Care Robots*	*Service Robots*	*Social Interaction Robots / Emotional Care Robots*
Concept of Proposed Technology	**Core Concept** The proposed technology is a culturally responsive Socially Assistive Robot (SAR) designed to support emotional stability, self-understanding, and self-efficacy among adolescents growing up in multicultural and migrant contexts, by providing a culturally safe and non-judgmental interaction environment. Beyond a simple emotion-responsive and empathetic dialogue system, this technology embeds the complex realities of migrant and multicultural environments into the core interaction design, including language mixing (code-switching), identity confusion, and communication disruptions across family and generational boundaries. In particular, the core concept of this technology lies in its design as a mediating structure in which interaction with the robot not only supports adolescents' individual emotional regulation, but also facilitates communication and connection with real human relationships, such as family members, teachers, and counselors.			

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Key Functions and Potential Applications

1) Bilingual and Code-Switching–Based Emotional Interaction
   - An LLM-based conversational engine that understands and generates mixed utterances (code-switching) naturally occurring in bilingual environments such as Korean–English.
   - By reflecting language-switching patterns according to context, emotional state, and topic, the system minimizes the cognitive and emotional burden of expression and enhances users' psychological comfort.
   - Provides a cultural and linguistic mediation function to help reduce communication barriers between parents and adolescents.
2) Embedded Evidence-Based Emotional Intervention Framework
   - Gradual integration of evidence-based mental health intervention techniques, including cognitive restructuring from Cognitive Behavioral Therapy (CBT) and Self-Compassion–based interventions, into robot interaction scenarios.
   - A structured dialogue flow—self-disclosure → emotion labeling → cognitive re-interpretation → real-world application practice—designed to progressively strengthen self-understanding and self-efficacy among adolescents.
3) Personalized Adaptive Interaction Policy
   - Dynamic adjustment of questioning strategies, conversational pace, empathy intensity, and level of intervention based on users' response patterns, emotional expression styles, and interaction histories.
   - Application of an adaptive interaction policy that learns individual emotional response characteristics through long-term interaction.
4) Real-Time Multimodal Empathic Behavior Generation
   - Real-time generation of empathic behaviors grounded in active listening, through integrated analysis of vocal prosody, speech rate, silence, gaze, and head movements.
   - Enhancement of emotional co-presence by modulating response timing and behavioral intensity in accordance with conversational context and emotional state.
5) Mediation Mechanism for Facilitating Human Relationships
   - Provision of a neutral and non-diagnostic summarization and communication support mechanism, enabling emotional insights and

	communication experiences formed through robot interaction to extend to real human relationships, such as family members, teachers, and counselors. ● Through this mechanism, the robot functions not as a replacement for human relationships, but as a technological mediator that supports relationship restoration and communication facilitation. -
Rationale for Joint R&D Project	**Critical Technical Barriers** As of 2024, the global migrant population has exceeded 300 million, and mental health challenges among adolescents growing up in migrant and multicultural environments have emerged as a major global public health and education issue, rather than a problem confined to any single country or group. In particular, adolescents from migrant or multicultural backgrounds are more likely to experience identity confusion, weakened sense of belonging, and emotional stress as they navigate between: ● Cultural and linguistic norms within the family ● Social norms and expectations required in schools and local communities. Despite often demonstrating adequate academic performance or outward adaptation, many of these adolescents continue to experience emotional difficulties. However, they face structural barriers that prevent effective connection to formal mental health support resources, including: ● Stigma associated with mental health services, ● Language barriers between parents and children ● Existing counseling systems that insufficiently reflect cultural and contextual factors. To address these limitations, this project proposes to develop a culturally responsive robot that acts as a non-judgmental and empathetic companion and provides a culturally safe interaction environment for adolescents with immigrant/multi-cultural backgrounds to self-express and navigate their emotional well-being. **Why Global Collaboration?** This project constitutes a convergent research effort that aims to technologically model the emotional experiences of adolescents growing up in linguistic and cultural transition environments and to validate these models in real-world educational and welfare settings. Due to its scope and complexity, the project faces structural limitations if conducted by a single institution or within a single national context. Emotional and communication challenges faced by adolescents from migrant backgrounds vary

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	significantly depending on national education systems, cultural norms, and immigration contexts. Therefore, international collaborative research spanning diverse cultural and linguistic environments is essential to ensure the generality and reproducibility of the proposed technology. This joint R&D project establishes an integrated research structure in which problem definition, technology design, and field validation are organically linked, through a clear division of roles between: ● An overseas research institute with expertise in education, language studies, and human–robot interaction (HRI) for multicultural and migrant youth, and ● Korean SME equipped with capabilities in the design, development, validation, and productization of emotionally supportive socially assistive robots. The overseas research institute will lead the theoretical refinement of culturally and linguistically grounded emotional perception and interaction models, as well as international comparative studies, while the Korean SME will be responsible for robot hardware–software integration, implementation of empathic interaction technologies, and enhancement of technical maturity through real-world deployment. Through this collaborative R&D framework, the project will secure both research rigor and empirical reliability, while simultaneously establishing a technology foundation capable of global application and scalability. Accordingly, international joint R&D is not an optional approach but a necessary research strategy for the successful execution of this project.
Key Words	1) Culturally Adaptive Socially Assistive Robot 2) Empathic Human–Robot Interaction (HRI) 3) LLM-based Bilingual & Code-Switching Conversational AI 4) Evidence-based Mental Health Intervention in HRI 5) Multimodal Empathic Behavior Generation

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	**Objective 1. Development of a Bilingual Dialogue Engine Specialized for Adolescents from Multicultural and Migrant Backgrounds** ● Develop an LLM-based conversational engine capable of understanding and generating adolescents' real-world language use patterns in bilingual and code-switching environments (e.g., Korean–English). ● Implement capabilities for conversation context retention, emotional signal recognition, and topic shifting, while incorporating safety guardrails for sensitive topics such as depression, self-harm, and crisis situations. ● Structure conversation logs and metadata to establish a foundation for personalized interaction and emotional response analysis. **Objective 2. Development of an Interaction Scenario Package Embedding Evidence-Based Emotional Interventions** ● Gradually integrate cognitive restructuring techniques from Cognitive Behavioral Therapy (CBT) and Self-Compassion–based intervention methods into robot interaction scenarios. ● Design a structured dialogue flow progressing from self-disclosure → emotion labeling → cognitive re-interpretation → real-world application practice, with the goal of enhancing adolescents' self-understanding and self-efficacy. ● Build a scenario library reflecting contexts frequently observed in multicultural and migrant settings, including identity confusion, achievement pressure, and family communication challenges. **Objective 3. Implementation of Real-Time Multimodal Empathic Behavior Generation and Active Listening Policies** ● Implement multimodal behavior generation technology that produces empathic behaviors in real time through integrated analysis of vocal prosody, speech rate, silence, spoken content, and nonverbal signals. ● Develop active listening–based behavior policies that regulate question timing, response intensity, and level of intervention, thereby improving interaction quality and emotional co-presence. **Objective 4. Establishment of Personalized Adaptive Interaction Policies and Long-Term Engagement Structure** ● Develop personalized adaptive interaction policies that adjust questioning strategies, language selection, empathetic responses, and depth of intervention based on each individual behavioral patterns and interaction histories. ● Establish a youth-friendly interaction structure that learns emotional response characteristics over long-term use and promotes sustained engagement. **Objective 5. Development and Empirical Validation of HRI-Based Human Relationship Facilitation Mechanisms**			

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- Design a triadic adolescent-parent-robot interaction paradigm in which the robot act as a communication facilitator between a parent and their child to improved communication and relationships. This interaction paradigm may apply to other relevant adults in the youths' lives, including family members, teachers, and counselors.
- Conduct pilot field studies in multicultural environments to quantitatively and qualitatively validate technical feasibility, user acceptance, and emotional impact.

R&D Target Specifications

Category	Performance Indicator	Target Level
Bilingual Dialogue Performance	Accuracy of understanding Korean–English mixed utterances	≥ 90%
	Dialogue context retention accuracy	≥ 85%
Quality of Empathic Responses	Appropriateness of empathic responses	≥ 85%
	Appropriateness of active listening behaviors	≥ 80%
Emotional Effectiveness	Change in self-efficacy	Effect size d ≥ 0.5
	Improvement in emotional stability (e.g., measured via PHQ-9, GAD-7)	Statistical significance achieved ($p < 0.05$)
Personalization & Sustained Use	Retention rate over 4 weeks	≥ 70%
	Satisfaction with personalized responses	≥ 80%
Multimodal Behavior Performance	Empathic behavior timing error	≤ 500 ms
	Naturalness score of nonverbal behaviors	≥ 80%
Safety and Ethical Compliance	Accuracy of responses to crisis and sensitive situations	≥ 95%
	Compliance with non-diagnostic and non-judgmental principles	100%

R&D Plan

This project aims to systematically advance technology maturity through a phased research approach consisting of core technology development → system integration and pilot validation → scaled validation and advancement.

1) Phase 1 (Core Technology Design and Development)
   - Establish conceptual models of emotional recognition and empathic interaction that reflect the linguistic and cultural transition environments experienced by adolescents from multicultural and migrant backgrounds.
   - Develop the core algorithms required for implementation, including multimodal emotion recognition, empathic response generation, and multilingual/code-switching–aware dialogue management technologies.
   - Conduct laboratory-based baseline performance validation to ensure technical feasibility.
2) Phase 2 (System Integration and Pilot Validation)
   - Integrate the core technologies developed in Phase 1 into a socially assistive robot platform.
   - Conduct pilot field deployments primarily in school and community settings.

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	● Evaluate key performance indicators, including facilitation of emotional expression, appropriateness of empathic responses, and transfer effects to human relationships. ● Identify technical and operational improvements required for real-world deployment based on pilot results. 3) Phase 3 (Scaled Validation and Technology Advancement) ● Conduct multi-site validation studies in both domestic and international settings to verify reproducibility and cultural applicability of the technology. ● Evaluate sustained usability and reliability in long-term use environments. ● Perform international comparative validations to assess global scalability and potential for broader dissemination.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Establish a theoretical model of emotional recognition and empathic interaction that reflects the linguistic and cultural transition environments experienced by adolescents from multicultural and migrant backgrounds, and present corresponding research directions for human–robot interaction (HRI). ● Lead the design of interaction scenarios and dialogue policies that account for multilingual and code-switching contexts, providing theoretical and methodological foundations for culturally grounded empathic interaction. ● Develop an empirically driven evaluation framework, including HRI-based experimental design, definition of quantitative and qualitative evaluation metrics, and analytical methodologies, to ensure objectivity and reproducibility of the research. ● Establish and manage validation protocols that strictly adhere to research ethics (IRB compliance), data privacy protection, and emotional safety standards throughout the research process, thereby ensuring international-level research reliability and accountability.
Expected R&R of Korean SME Partner	● Design the integrated hardware–software architecture of the emotionally supportive socially assistive robot and oversee system development to ensure the stable implementation and operation of core AI interaction technologies. ● Develop and advance core AI interaction technologies—including multimodal emotion recognition, empathic response generation, and dialogue flow control—to a level suitable for real-world deployment. ● Apply the multilingual and code-switching–capable dialogue engine to actual usage environments, and secure operational quality through performance optimization focused on response latency, accuracy, and system stability. ● Technically implement, test, and iteratively refine robot interaction scenarios based on the interaction theory models and scenarios designed by the overseas research institute. ● Operate pilot field deployments in real-world environments such as schools and community settings, and continuously enhance technical maturity through the collection and analysis of user logs and system data.

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	● Ensure safety and reliability across technical and operational domains, including robot hardware stability, data management systems, and personal data protection. ● Validate productization feasibility based on field validation results, and lead follow-on commercialization and market entry by designing B2B and B2G–oriented business models and dissemination strategies.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	● Secure core technologies for emotion-recognition–based socially assistive robots that reflect the linguistic and cultural transition environments of adolescents from multicultural and migrant backgrounds, and empirically validate the differentiated emotional interaction technologies, including multimodal emotion recognition, empathic response generation, and multilingual dialogue management. ● Propose a relationship-transfer emotional care model that extends human–robot interaction (HRI) to human–human interaction (HHI), demonstrating that robot-based emotional support can lead to enhancement in interpersonal cross-cultural relationships and communication. ● Verify the effectiveness of preventive and supportive emotional care service models and their applicability in public education and welfare settings through real-world field validations conducted in schools and community environments. ● Establish application guidelines encompassing research ethics, personal data protection, and emotional safety standards, and validate the technology's reproducibility, cultural applicability, and international scalability through domestic and international joint validations. ● Based on field validation outcomes, build a foundation for productization and B2B/B2G-oriented commercialization, and present clear pathways for service expansion across education, welfare, and public sectors, as well as entry into global markets.

6. References

Migration Policy Institute (MPI). (2024). ***Top Statistics on Global Migration and Migrants, 2024.***